

qSOLIS

User Manual

Version 1.0.0

Solar visibility, irradiance, and irradiation analysis for CloudCompare

CloudCompare 2.14+ / Qt 6
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1. Overview

qSOLIS evaluates solar exposure on CloudCompare point clouds and triangular meshes. It combines a solar-position/clear-sky irradiance model with an offscreen OpenGL visibility pass. Results are written as CloudCompare scalar fields on the original point or vertex cloud.

- Direct solar visibility or irradiance.
- Diffuse sky visibility or irradiance.
- Total irradiance/irradiation as Direct + Diffuse.
- Instantaneous irradiance in W/m2.
- Mean irradiance over an interval in W/m2.
- Integrated irradiation in Wh/m2 or kWh/m2.
- Horizontal-reference, true surface (normal-aware), and visibility modes.
- GUI and command-line workflows using the same numerical engine.

2. Coordinate and time conventions

qSOLIS assumes +X = east, +Y = north, and +Z = up. Longitude is east-positive; latitude is positive north and negative south.

Day-of-year (DOY) runs from 1 through 366. DOY 0 is invalid. A clock time is stored internally as the fractional part of the day-of-year.

Time can be entered as UTC or as local civil time with a fixed UTC offset. The local-time mode is a fixed offset only; it does not apply named timezone or daylight-saving rules automatically.

3. Starting a calculation

1. Load a point cloud or mesh.
2. Select the point cloud itself or the mesh in the DB tree.
3. Choose Plugins > SOLIS Calculate....
4. Configure the calculation.
5. Press OK.

The dialog displays qSOLIS v1.0.0 at the bottom so calculation screenshots and reports can identify the plugin version.

3.1 Location and time

Set latitude, longitude, elevation, DOY, time, and time reference. For UTC, leave the UTC offset disabled. For local fixed-offset time, choose Local and enter the offset from UTC.

3.2 Output quantity

Mode	Output	Meaning
Instantaneous	W/m2	One time point
Mean	W/m2	Average over configured interval
Integrated energy	Wh/m2 or kWh/m2	Time-integrated irradiation

For Mean and Energy modes, Duration and Timestep become active.

3.3 Spatial mode

Mode	Use
Horizontal reference	Visible points receive horizontal-plane incidence weighting.
True surface	Uses each point/vertex normal; normals are required.
Visibility	Returns visibility fractions and ignores irradiance magnitude.

If any selected point cloud or mesh vertex cloud lacks normals, Surface mode is disabled. In Visibility mode, Total is undefined; Direct + Diffuse creates two separate visibility fields.

3.4 Components

Select Direct, Diffuse, or both. When both are selected in an irradiance mode, qSOLIS also creates Total = Direct + Diffuse point-by-point.

3.5 Quality presets

Preset	Render resolution	Diffuse rays
Fast	512	64
Normal	1024	256
High	2048	512
Custom	User selected	User selected

Higher resolution improves shadow-boundary detail but increases GPU memory, framebuffer readback, and CPU projection work. Normal is the recommended starting point.

4. Advanced options

Expand Advanced options for controls that are normally left at defaults.

- Linke turbidity: clear-sky atmospheric attenuation; one constant value per calculation.
- Resolution: square visibility framebuffer size.
- Depth bias: depth-comparison tolerance for visibility.
- CPU threads: Auto, 1 for single-threaded, or an explicit worker count.
- Closed mesh optimization: enable only for known watertight meshes.
- Diffuse rays: deterministic hemisphere sample count.
- Direct angular grouping: optional approximation that merges nearby sun directions for long integrations.

5. Batch time series

Enable Batch time series to store multiple calculation start times as separate scalar fields without duplicating geometry.

Set an end DOY/time and a step in minutes. For Mean or Energy modes, each batch timestamp is the start of its own interval. The GUI asks for confirmation for large batches and qSOLIS limits one batch to 1000 time points.

6. Result layers and naming

Each persistent result is a new scalar field on the original point/vertex cloud. Existing qSOLIS results are never silently overwritten.

```
SOLIS Direct [W/m2] D175 T1200 UTC H R1024
SOLIS Diffuse [W/m2] D175 T1200 UTC H R1024
SOLIS Total [Wh/m2] D175 T0000 UTC dur24h dt5m H R1024
SOLIS Direct Mean [W/m2] D175 T1200 UTC dur2h dt5m S R2048
```

Duplicate names receive #2, #3, and so on. Full calculation provenance is stored as qSOLIS JSON metadata, including plugin version, run/batch identifiers, location/time, atmosphere, rendering, and sampling settings.

7. Cleanup

Select one or more clouds/meshes and choose Plugins > SOLIS Cleanup....

- Last run.
- Last batch.
- Selected batch.
- All metadata-backed qSOLIS results.
- Explicit legacy SOLIS-name cleanup.

Normal cleanup is metadata-driven and leaves unrelated scalar fields untouched. Legacy cleanup is intentionally separate because it relies on name matching.

8. Command-line use

qSOLIS registers -SOLIS and -SOLIS_CLEAN. The complete option reference is in docs/CLI.md.

```
CloudCompare -SILENT -AUTO_SAVE OFF -O tree.ply -SOLIS \
  -COMPONENT DIRECT -OUTPUT_MODE INSTANT -SPATIAL_MODE HORIZONTAL \
  -DOY 172 -TIME 12:00 -LAT 45 -LON 0 -ELV 0 -RESOLUTION 1024
```

Every created field is printed to the CLI log. Automation should capture these exact names rather than reconstructing duplicate suffixes.

9. Interpretation of results

- Direct: clear-sky direct normal irradiance, visibility, and the selected incidence term.
- Diffuse: isotropic sky radiance sampled over a deterministic hemisphere; Surface mode applies normal incidence.
- Total: Direct + Diffuse for irradiance/energy modes.
- Direct visibility over an interval: fraction of sampled daylight time visible to the sun.
- Diffuse visibility: fraction of sampled sky solid angle visible from the point/vertex.

10. Performance guidance

1. Start at Normal quality (1024 / 256 rays).
2. Compare 1024 and 2048 before adopting High for production.
3. Use a coarser temporal timestep when scientifically acceptable.
4. Benchmark direct angular grouping against an ungrouped reference before relying on it.
5. Keep CPU threads on Auto unless benchmarking indicates otherwise.

qSOLIS reuses uploaded geometry and the rendering backend across Direct/Diffuse and batch points.

11. Regression tests

The release contains qSolis/tests/regression with surface-normal, missing-normal, and wall-shadow acceptance cases.

```
cd qSolis/tests/regression
./run_qsolis_regression.sh /path/to/CloudCompare
```

Frozen v1.0.0 result	Value
Surface horizontal median	262.5244 W/m2
Surface south median	665.6770 W/m2
Surface north median	0.0000 W/m2
45 deg south/up median	656.6156 W/m2
Wall sunlit median	834.6073 W/m2
Wall shadow length	1.9746 m

12. Troubleshooting

Symptom	Check
SOLIS action disabled	Select the actual point cloud or mesh, not only a parent container.
Surface mode disabled	Compute/import normals for every selected point/vertex cloud.
All-zero direct result	Verify sun above horizon and check location/time; inspect OpenGL errors if resolution-specific.
CLI on headless host fails	-SILENT removes dialogs but not the Qt/OpenGL requirement.
Too many result fields	Use SOLIS Cleanup > Last batch.

13. Scientific limitations

- No measured cloud cover or meteorological time-series input.
- Isotropic diffuse-sky model.
- Constant Linke turbidity over one calculation.
- Surface results depend on normal quality and orientation.
- Shadow boundaries depend on finite render resolution and depth bias.
- Day-of-year solar approximations do not carry an explicit calendar year.

Document these assumptions when qSOLIS results are used in research or engineering reports.

14. Version and license

This manual describes qSOLIS v1.0.0, released 2026-10-15.

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